



Indian Institute of Information Technology Surat
Department of Computer Science and Engineering
End Semester Examination, 3rd Sem, 2025-26
Automata and Formal Languages (CS304)

Timing: 10AM-1PM Date: 9 Dec 2025 Maximum Marks: 100

Instructions: This question paper is printed on **both sides**. Ensure that your solutions **follow all the conventions** discussed in the class. Attempt all parts of a question together. Attempt only the required number of sub-parts in a question. If you do not, later parts may not be graded.

Question 1. Our class representative Parth is revising for the end-semester exams but has the following misconceptions. Correct each one of them (without explanation) by writing the intended correct statement using standard terminology.

- Every regular language has a unique regular expression that represents it. (2)
- The language of the names of all current students at IIIT Surat is irregular. (2)
- The full name of the class NP is Non-Polynomial time. (2)
- Every context-free language is regular. (2)
- The statement "This Statement is False" is true. (2)

Question 2. Consider the language L over $\Sigma = \{a, b\}$ consisting of all strings that contain at least one b . **Attempt any 3.**

- Construct an NFA that accepts L . (5)
- Write a regular expression for L . (5)
- Write a CFG for L . (5)
- Define the complexity class P . Give a brief description as to why $L \in P$. (5)

Question 3. For each of the following, write in a box whether it is **True** or **False** or **Undecidable** or an **Open Question**. Justify your answer with either a brief explanation (**maximum 3 sentences**) or a counterexample. **Attempt any 3.**

- There exists a TM, that takes input of a description of a CFG G , and then overwrites the input tape with the description of a CFG G' such that $L(G) = L(G')$ and G' contains no useless symbols. (5)
- PRIME is in P . (5)

c. Consider the problem LOOP defined as follows: Given a TM M and an input string w , does M loop on w ? Then the problem LOOP is recognizable. (5) ✓

d. For languages A and B with $A \leq_p B$, if $A \in NP$ and $B \in P$, then $P = NP$. (5)

Question 4. Consider language $L = \{01, 0011, 000111, 00001111, \dots\}$ over $\Sigma = \{0, 1\}$.

a. Write the recursive definition for L . Clearly state the base case, the recursive step, and the closure. (5)

b. State (without proof) the pumping lemma for regular languages. Prove that L is not regular using the pumping lemma. (10)

c. Give a brief definition of context-free languages. Prove that L is context-free by providing a context-free grammar for it. (5)

Question 5. For each of the following, write in a box whether it is True or False. Justify your answer with a brief explanation (**maximum 3 sentences**) or a counterexample. Attempt any 3.

a. Let NFA_1 be the class of NFAs with exactly one accept state. Then, NFA_1 is less powerful than NFA. (5)

b. For every regular language, there exists a unique minimal DFA that accepts it. (5)

c. The intersection of a regular and a context-free language is context-free. (5)

d. If a CFG is in Chomsky Normal Form, then it is unambiguous. (5)

Question 6. Recall that a Hamiltonian path in a graph is a path that visits each vertex exactly once and a Hamiltonian cycle is a cycle that visits each vertex exactly once. Consider the problem HAMPATH: given a graph G and two vertices s and t , does there exist a Hamiltonian path from s to t in G ? Also consider the problem HAMCYCLE: given a graph G , does there exist a Hamiltonian cycle in G ? Answer the following:

a. Define the complexity class NP. What TM should you construct to prove that $HAMPATH \in NP$? Give a very high level overview of this TM. (5) ✓

b. Prove that if $HAMCYCLE \in P$ then $HAMPATH \in P$. (15) ✓

c. To prove that HAMCYCLE is NP-Complete, we need to prove that $A \leq_p B$ for some problems A and B . What are A and B ? Justify your answer in 1 sentence. (5)